

SECTION B

6 (**short**) Using Laplace transforms, solve the differential equation

$$\frac{d^2x}{dt^2} - 4x = \sinh t$$

with initial conditions $x = dx/dt = 0$ at $t = 0$. [10]

7 (**long**) A linear system is described by

$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + 5y = f(t)$$

with initial conditions $y = dy/dt = 0$ at $t = 0$.

(a) Show that the step response is

$$y(t) = \frac{1}{5} - e^{-t} \left[\frac{\cos 2t}{5} + \frac{\sin 2t}{10} \right]$$

and find the impulse response. [12]

(b) Using convolution, determine the system response to the input

$$f(t) = \begin{cases} e^{-\alpha t} & t \geq 0 \\ 0 & t < 0 \end{cases}$$

where α is a positive constant. What is the significance of the limit $\alpha \rightarrow 0$? Comment briefly on your result. [18]